

Module-1

Lecture-6 Strain Energy

MECHANICS OF SOLIDS

- Strain Energy
- Strain energy in different types of loading
- Problems

Introduction

Strain Energy

- Energy absorbed in body when strained within its elastic limit
- Strain energy is capable of doing work

$$\text{Strain Energy} = \text{Work done}$$

Resilience

- Total strain energy stored in a body
- Capacity of strained body to do work

Proof Resilience

Max strain energy which can be stored in a body

Modulus of resilience

Proof resilience per unit volume